

Data curation and hyperparameter transfer enable competitive variant-effect prediction with a 1B genomic language model

TODO

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Abstract

Genomic language models can learn functional constraints directly from DNA sequence, but strong performance has often depended on specialized architectures, very large training budgets, or information derived from alignments and functional genomics. We developed MarinDNA to test how far a standard decoder-only Transformer can be taken through functional-region data curation, explicit mixture design, and systematic hyperparameter transfer. Hyperparameters calibrated on approximately 25M-parameter reference models identified the best observed learning rate at 255M, 476M, and 1B parameters, and the transferred recipe produced predictable loss scaling across eight models from 46M to 4B parameters. Continued training on a five-region mixture yielded a 1B model trained on 166B tokens that was statistically competitive with Evo 2 40B on the frozen Mendelian variant-effect benchmark. Relative to Evo 2 40B, this model used approximately 1,980-fold fewer reported training FLOPs and achieved approximately 2,330-fold higher as-deployed scoring throughput at each model’s native context length, although alignment-based and functionally supervised models remained stronger overall. These results show that careful data and optimization choices can make a conventional, short-context genomic language model competitive at substantially lower training and deployment cost.

1. Introduction

Optimization of genomic language models (gLMs) has historically involved a lot of focus on specialized architectures.¹ We take a different approach: keep the architecture within the standard GPT family and focus on data curation and mixtures, training hyperparameters, and model scale. The experiments below test how far that simpler, standardized recipe can go.

1.1. Why alignment-free gLMs?

Many of the strongest genomic sequence models rely on whole-genome alignments, as in [GPN-Star](#),² or functional-genomics measurements, as in [AlphaGenome](#)—resources concentrated in a small number of well-studied species because they are costly to produce and difficult to construct comprehensively. Unlabeled DNA sequence, by contrast, is available for a rapidly growing number of species. Alignment-free gLMs—also called single-sequence gLMs—can be trained directly on this growing collection of genomes and applied to evolutionary constraint prediction, sequence design, and transfer learning.

For humans and other well-studied species, single-sequence gLMs are still far from replacing alignment-based models or models supervised with functional-genomics data. Our near-term goal is to build useful models for species that lack high-quality whole-genome alignments and functional-genomics data. As a concrete example, we aim to enable accurate, cost-effective genome-wide mapping of evolutionary constraint for any mammalian

¹For a broader overview, see “[Genomic language models: opportunities and challenges](#)”. Examples of specialized architectures include long-convolution or hybrid long-context models such as [HyenaDNA](#) and [Evo 2](#); U-Net-like models such as [NTv3](#); bidirectional models such as [DNABERT-2](#), [GENA-LM](#), [PlantCAD2](#), and [TrinityDNA](#); state-space or hybrid state-space models such as [HybriDNA](#), [Caduceus](#), and [PlantCAD2](#); reverse-complement equivariance in [Caduceus](#); learned adaptive segmentation that replaces fixed tokenization in [dnaHNet](#) and [DNACHUNKER](#); the hierarchical, multiscale Transformer used by [megaDNA](#); and early or less-established sparse-expert models such as [JanusDNA](#), [PlantBiMoE](#), and [MxDNA](#).

²Exact details are not yet available, but [Omnii](#) also appears to benefit from alignment-derived information.

³Zoonomia illustrates the gap between having an alignment and having a ready-to-use constraint track for each mammalian genome. Producing these tracks requires running per-base scoring in each target genome’s coordinate system, which can require substantial

species.³ Longer term, sequence-only models may learn from sequence in ways that complement alignments, conservation scores, and functional-genomics models even in data-rich species ([research question #396](#)).

1.2. Why data curation?

Many early gLMs were trained on the human genome alone, with little to no filtering. Subsequent work made it clear that two factors were key drivers of model performance: including multiple species, and enriching for functional regions rather than sampling uniformly from the majority-neutral background of mammalian genomes.⁴

In this work, we follow up on two findings from [TraitGym](#). First, 152M-parameter GPN-Promoter, trained only on animal promoters, performed comparably to Evo 2 40B on human promoter variants. Second, Evo 2 improved substantially with scale overall but still struggled on distal variants. Enhancers were the only region type not actively curated into its training data and were sparse among the intergenic regions it saw.

MarinDNA therefore treats dataset construction as a primary modeling lever: which species and evolutionary timescales to include ([research question #394](#)), which functional regions to sample and how to identify them ([research question #395](#)), how to weigh them, and when during training to introduce them.

We began with annotation-derived datasets for coding, upstream, and downstream sequences.⁵ Standard genome annotations make these regions relatively easy to identify and extract consistently across many species.

Later, we added ncRNA⁶ and enhancers⁷ built by alignment projection. Because comparable annotations were not directly available across the target species, we projected human annotations through whole-genome alignments. The broader idea of using whole-genome alignments to expand reference-species enhancer data across species has also appeared recently in chromatin-accessibility prediction and enhancer-generation work.⁸

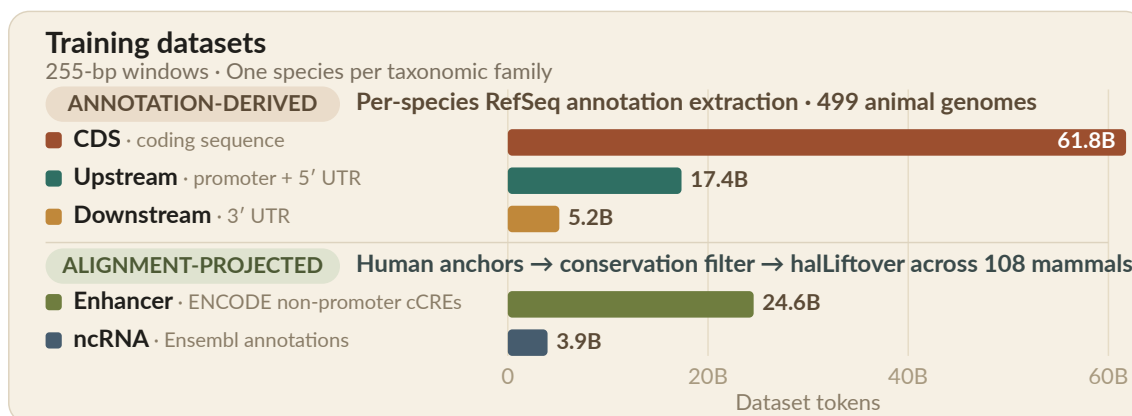


Figure 1 Training-dataset provenance and scale.

1.3. Why GPT-style architecture?

By GPT-style, we mean the deliberately simple approach of treating DNA as text and training a stock causal, autoregressive, decoder-only language model on it. In these experiments, that architecture is literally Qwen3 rather than a genomics-specific design. This approach is not new; a substantial line of prior gLM work has used causal

intermediate storage and compute. The project’s [data page](#) links to a reference-free 241-way mammalian alignment, but the associated [CGL resource page](#) offers a single score download explicitly labeled “Human PhyloP scores.” The [expanded 447-way resource](#) publishes alignment files but no phyloP score downloads; UCSC’s standard 447-way [phyloP track](#) is likewise exposed only under human hg38. This is part of a broader gap: as of July 2026, UCSC’s standard [goldenPath](#) download tree contained phyloP tracks for only 20 species, nine of them mammals.

⁴See *Genomic language models: opportunities and challenges*, GPN, PlantCaduceus, PlantCAD2, Shorkie, GPN-MSA, Species-aware DNA language models, nucleotide-dependency analysis, Evo 2, and ARSENAL.

⁵“Downstream” denotes the 256 bp immediately downstream of each annotated CDS end, rather than annotated 3' UTR intervals. In [experiment #53](#), this distance-based definition produced better 3' UTR VEP performance than the annotation-derived baseline.

⁶Most species had ncRNA annotations, but in [experiment #43](#), an annotation-derived ncRNA specialist showed little improvement on ncRNA variants. This motivated the later use of human annotations projected through whole-genome alignments.

⁷“Enhancer” is shorthand for the ENCODE V4 non-promoter cCRE set. It includes enhancer-like signatures (dELS and pELS), but also classes such as CA, CA-CTCF, CA-TF, CA-H3K4me3, and TF.

⁸STEAM augments mouse chromatin-accessibility training data with syntenic orthologs from 240 other mammalian genomes, while EnhancAR trains an autoregressive generator on families of human enhancers and their homologs.

language modeling with GPT- or Llama-like architectures.⁹ What is new here is the quality target. Even recent models in this family, such as Carbon, generally aim for non-inferiority to smaller Evo 2 checkpoints and still underperform Evo 2 40B on the broad zero-shot VEP setting we care about.¹⁰ If the quality gap can be closed, GPT-style models have obvious advantages for deployment. They run through familiar training and inference stacks, move cleanly across hardware, and avoid model-specific kernels or bespoke architecture code, which matters a lot for cost, flexibility, and usability.

MarinDNA’s first experiment compared three training objectives on promoter sequences: causal language modeling, masked language modeling, and masked diffusion ([experiment #3](#)). Causal language modeling looked most promising in the initial training steps. This was not a definitive matched-compute comparison of objectives, but it provided enough direction-setting evidence to pursue a simple causal architecture. That choice also need not permanently constrain the model to left-to-right representations: decoder-only language models can be adapted into bidirectional encoders with further training ([research question #393](#)).

Our initial goal was to model individual functional elements of the genome—such as an exon or an enhancer—well. We therefore deliberately used a short 255-bp context: it was sufficient for the functional-element tasks we tested and made training and evaluation much faster. Centering each example on an individual functional element also made the training data easier to construct, filter, audit, and interpret. How best to extend context—either through additional long-context next-token pretraining or during downstream-task fine-tuning—remains future work ([research question #392](#)).

1.4. Why VEP evaluation?

Variant effect prediction (VEP) is one of the most important applications of gLMs. A useful VEP model can help scale clinical interpretation for rare disease, hereditary cancer, and variants of uncertain significance.¹¹ It can also help connect genetic association signals to disease mechanisms, target selection, and causal-variant prioritization in GWAS fine-mapping.¹² The same kind of evidence is relevant to clinical trial design when genetics can inform patient stratification, enrollment criteria, or mechanism-based cohort definition. Together, these are commonly used levers for improving the efficiency of pharmaceutical development, and it is uncommon for other gLM evaluations to have such a direct connection to commercially relevant research tasks. VEP is also one of the few evaluations backed by decades of costly clinical genetics curation, with resources such as ClinVar and OMIM providing a level of human variant evidence that has no real analogue in other species.¹³ That combination makes it a substantive test of whether a gLM has learned sequence constraints that actually matter for human biology. If a model has learned useful sequence-level constraints from DNA alone, it should help rank variants in places where direct experimental evidence is weak or nonexistent.

This creates a deliberate mismatch between evaluation and intended use: although we expect single-sequence gLMs to be especially valuable for non-model organisms in the near term, we evaluate them in humans because comparable variant-effect data are not yet available across species. Human VEP is therefore the most rigorous available test of learned functional constraint, but it does not by itself establish transfer to other organisms; evaluating that transfer will require broader population-genetic or experimental datasets.

⁹GPT-style or otherwise causal genomic models include [GenSLM](#), [DNAGPT](#), [LOL-EVE](#), [METAGENE-1](#), [GENERATOR](#), [GENERATOR-v2](#), [Gene42](#), and [Carbon](#). The closest human-DNA precedents are Carbon, GENERATOR, Gene42, DNAGPT, and the promoter-focused LOL-EVE; several of the others are important causal gLM examples but are less directly relevant to human VEP.

¹⁰The [Carbon-3B model card](#) describes Carbon-3B as a 3B-parameter decoder-only autoregressive genomic model implemented as a stock LlamaForCausalLM, with 6-mer DNA tokenization, long-context support, and a two-stage training schedule that switches from a standard cross-entropy objective to a factorized nucleotide supervision loss, bridging its coarse 6-mer tokenization with single-nucleotide resolution. Its public zero-shot table compares to Evo 2 7B, not Evo 2 40B: Carbon-3B is slightly ahead on BRCA2 and ClinVar noncoding, but behind on ClinVar coding and TraitGym Mendelian.

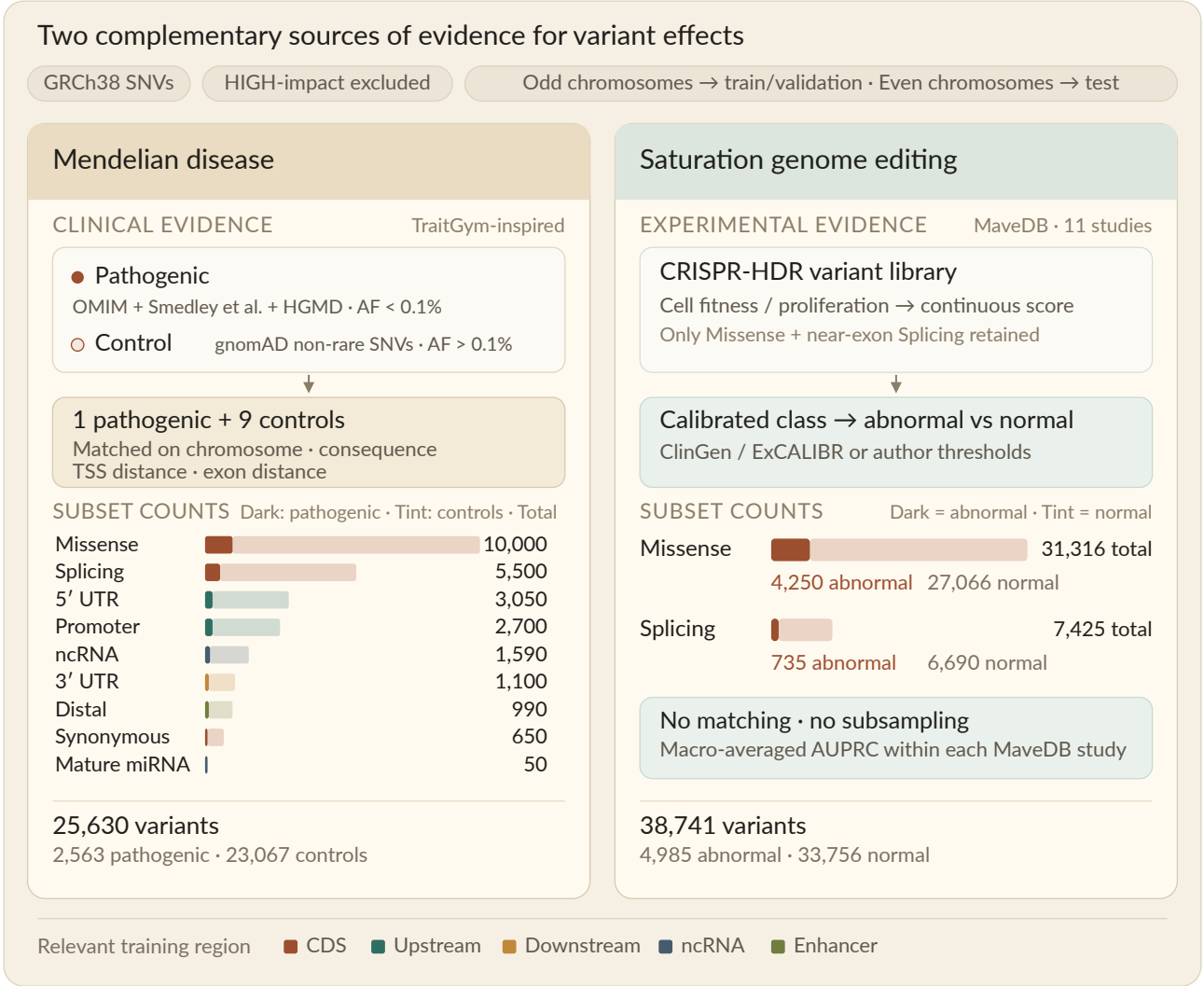
¹¹Examples include zero-shot or disease-focused variant interpretation results in [Evo 2](#), [GPN-Star](#), [Carbon](#), and [EnTao-GPM](#).

¹²Examples include fine-mapped GWAS and broader human-genetics results in [GPN-Star](#), regulatory variant-effect prediction in [AlphaGenome](#) and [ChromBPNet](#), and the broader observation that human genetic evidence can support target-disease hypotheses in drug discovery in [Nelson et al.](#)

¹³ClinVar archives submitted reports relating human genomic variation to disease, cancer, drug response, and supporting evidence; OMIM is a curated catalog of human genes, genetic disorders, and gene-phenotype relationships. Nothing comparable exists for any other species: this depth reflects decades of clinical genetics effort directed specifically at human disease, an investment that has simply not been made for non-human genomes.

In this work, we focus on predicting deleteriousness, pathogenicity, or, more generally, functional constraint. This task is the one most directly connected to the language-modeling training objective and is therefore easy to evaluate with zero-shot or linear-probing protocols. We leave the prediction of changes in gene expression—the main application of sequence-to-function models—to follow-up work, as it requires more complex fine-tuning protocols and much larger context sizes.¹⁴

We use two complementary sources of evidence: clinically curated Mendelian variants¹⁵ and saturation genome-editing (SGE) measurements. The Mendelian benchmark compares pathogenic and putatively benign variants across broad coding and non-coding consequence types. The SGE benchmark uses experimentally measured variant effects from a few genes in MaveDB, currently covering missense and splicing variants.



likelihood reflects functional constraint: deleterious alternate alleles should incur larger likelihood penalties relative to the reference allele. The probe instead asks what variant-relevant information is encoded in the model’s learned representation, including information that may not be directly reflected in its sequence likelihoods.

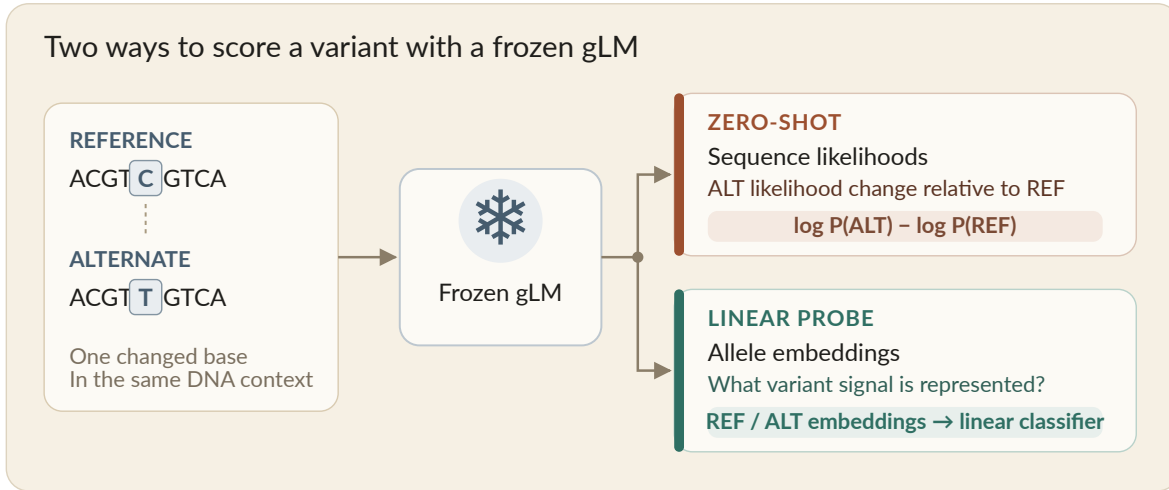


Figure 3 Variant effect prediction scoring approaches.

1.5. Why compare with Evo 2 40B?

Evo 2 40B remains the strongest broadly evaluated baseline among single-sequence DNA models. Despite substantial subsequent work on gLMs, we are not aware of a model in this setting that has clearly surpassed it across diverse genomic regions; many newer models do not attempt a direct comparison with Evo 2 40B. The other reason Evo 2 40B matters is its training budget. Its reported 2.25e24 training FLOPs are unrivaled among gLMs, corresponding to roughly \$2.5M of H100 time.¹⁶ That budget is unusual in biology and comparable to major open-weight LLM training runs from recent model generations,¹⁷ e.g. just above Qwen2.5-14B and below Qwen2.5-32B, and roughly between DeepSeek-V2 and DeepSeek-V3. Evo 2 40B is therefore the right baseline for asking whether a much simpler single-sequence gLM can be competitive.

2. Results

2.1. Early mixture experiments

We first trained a 1.7B upstream-region specialist, trying to replicate the success of GPN-Promoter ([experiment #21](#)).¹⁸ Although we used reasonable defaults rather than the systematic hyperparameter-transfer recipe developed later, performance was broadly comparable to Evo 2 40B. We saw a similar pattern when training a CDS specialist ([experiment #27](#)). Overall, however, GPN-Star remained stronger.

¹⁶This estimate uses the [Evo 2](#) reported training compute of 2.25e24 FLOPs, 50% H100 model FLOP utilization following the costing convention in [Beyond Chinchilla](#), 989 TFLOP/s BF16 peak throughput for an H100 SXM, and \$2 per H100-hour from [OLMo 3](#). The resulting accelerator requirement is about 1.26M H100-hours.

¹⁷Other over/under examples give the same intuition. The [AI2 OLMo 2 32B model card](#) places Evo 2 40B above Gemma 2 27B, OLMo 2 32B, and Llama 3.1 8B. A dense-accounting estimate from the [Llama 3.1 model card](#) places it well below Llama 3.1 70B.

¹⁸The upstream and CDS datasets used in these early experiments were earlier versions of those summarized in [Fig. 2](#): broadly comparable, but not identical. These models also used a 512-bp context without a BOS token, roughly twice the 255-bp context adopted for the later recipe.

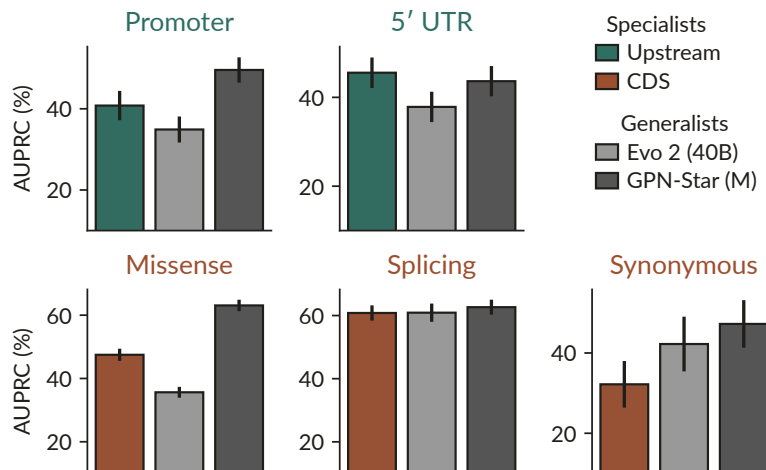


Figure 4 **Region-matched zero-shot Mendelian VEP: specialists versus generalists**. Error bars denote SE. After testing upstream and CDS specialists independently, the next experiment asked whether one model could retain both capabilities ([experiment #13](#)). Sampling in proportion to dataset size—10% upstream and 90% CDS—is the naive default when the two datasets are simply pooled without reweighting. In practice, it behaved similarly to CDS-only training. Equal 50/50 upstream/CDS sampling produced balanced performance across both regions. This made explicit mixture control a central axis of investigation. Even the 50/50 mixture may not be optimal: regions can differ both in size and in the density of learnable biological signal.

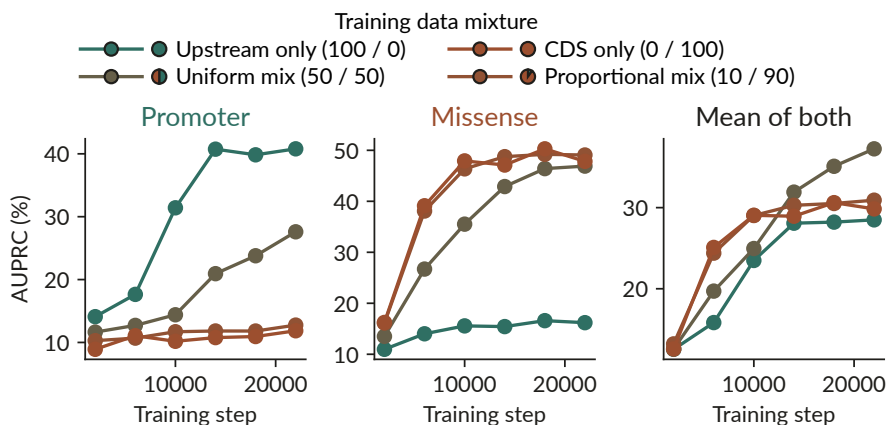


Figure 5 **Upstream/CDS training-mixture comparison**. Zero-shot results; the right panel is the unweighted mean of the promoter and missense AUPRC values.

2.2. Hyperparameter transfer

Our first attempt to scale manually lowered the learning rate as the model grew from 0.6B to 1.7B and 4B parameters ([experiment #57](#)). The larger models did not improve over the 0.6B model, and the early 4B run became unstable. That failure made systematic hyperparameter transfer a prerequisite: without a trustworthy optimization recipe, a model-size comparison would confound scale with tuning quality.

The annotation-derived DNA pool available at the time contained ~85B tokens. [Fig. 7](#) shows its proportional CDS, upstream, and downstream composition.

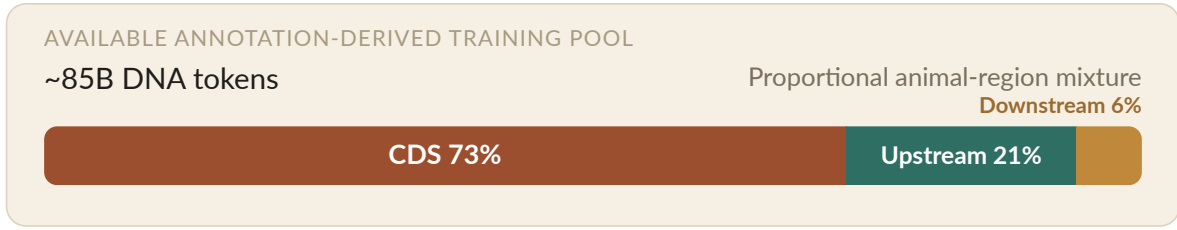


Figure 6 Available annotation-derived training pool.

That pool is large by genomics standards but small relative to modern accelerator-era training corpora. This makes the project data-constrained in principle even though our practical constraints are messier. We train on preemptible Google TPU Research Cloud resources, do not have consistent access to slices much larger than roughly 32 H100s worth of peak FLOPs, and want the recipe to remain reproducible at academic compute scale. O(100B) tokens therefore lands in an awkward middle ground where compute-constrained methods are still relevant, even though modest epoching is possible and likely breaks their assumptions at some unknown rate. We started with hyperparameter transfer for that reason. If a proven data-constrained transfer framework existed, we would use it. We do not know of one, so we followed the same basic pattern as [Delphi](#), fitting a small reference sweep with the Vizard Bayesian optimization framework and then scaling the result using a Complete(d)-inspired AdamH heuristic.¹⁹

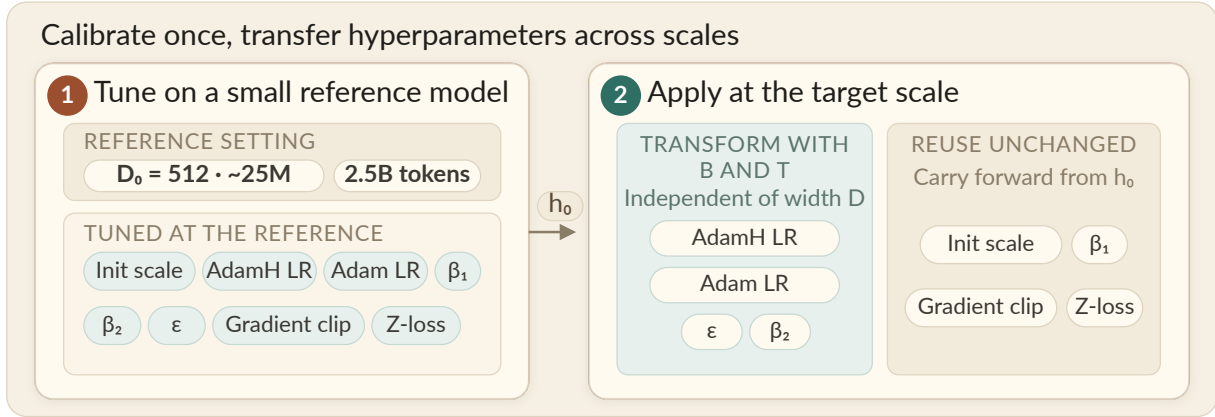


Figure 7 Hyperparameter calibration and transfer.

[Fig. 8](#) separates reference calibration from target application. Two heuristics sit behind that workflow and are fixed before tuning: an inherited rule maps hidden width D to model geometry,²⁰ and a second rule maps reference optimizer settings to a new batch size B and token horizon T . The reference sweep tunes initialization scale, the two learning rates, β_1 , β_2 , ϵ , gradient clipping, and z-loss. Here, tuning a learning rate means tuning its peak value under a fixed fractional schedule; the target run reuses that schedule shape, so warmup and decay scale with the run length rather than keeping fixed absolute step counts.²¹ At the target, the two learning rates, β_2 , and ϵ are transformed with B and T , while initialization scale, β_1 , gradient clipping, and z-loss are reused unchanged.²²

The reference sweep used ~25M-parameter models trained for 2.5B tokens with a 16k-token batch, or roughly 4e17 FLOPs per run. We then validated the transferred hyperparameters across 255M–1B-parameter models, with 4x as many tokens, 1/4x the batch size, and roughly 170x the FLOPs per run. The first test was whether the

¹⁹Complete(d) refers to the compute-constrained hyperparameter-transfer framework described in [Complete\(d\): Data-Optimizing Hyperparameter Transfer](#).

²⁰The DNA experiment inherits its geometry rule from Marin’s text-model scaling heuristic rather than fitting it on DNA. For hidden width D , it sets $\text{layers} = \text{round}(D/(55 + 4 \cdot \log_2 D))$, MLP width = $4D$, and both attention-head and KV-head counts to $D/128$; the sweep widths are selected manually. See the [commit-pinned geometry rule](#) and [model builder](#).

²¹Both learning rates use 10% linear warmup, remain at their peak through 80% of training, and then decay linearly to zero over the final 20%. These are fractions of the target run’s total steps. See the [experiment configuration](#) and [scheduler implementation](#).

²²Relative to the reference batch and token horizon (B_0 , T_0), the fixed heuristic uses AdamH LR $\propto \sqrt{(B/B_0) \cdot (T_0/T)} \cdot 0.3$, Adam LR $\propto \sqrt{(r/r_0)}$, $\epsilon \propto \sqrt{(r_0/r)}$, and $\beta_2 = \text{clip}(\beta_2, 0^{\wedge}(B/B_0))$, where $r/r_0 = (B \cdot T_0)/(B_0 \cdot T)$ and configured bounds still apply. The 0.3 exponent is inherited from Marin’s text recipe rather than fitted on the DNA sweep; the Complete(d) paper proposes a 0.5 token-horizon exponent, while a later [Marin text sweep](#) estimated ~0.28. See the [commit-pinned implementation](#).

learning-rate prediction survived that regime. Fig. 9 shows that the transferred prediction lands exactly on the best observed learning-rate setting at all three validation scales, outperforming both the unchanged reference optimum and every other target-scale sweep setting; the less sensitive optimizer hyperparameters are shown separately in Fig. 10.

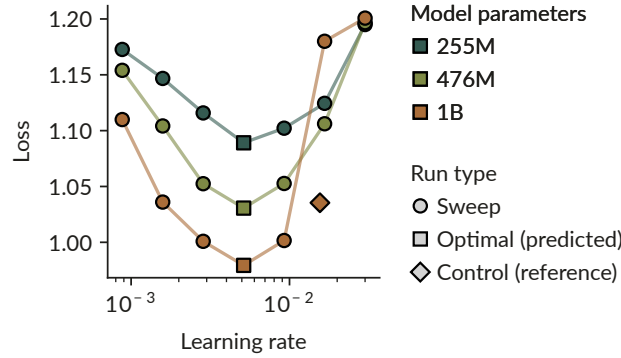


Figure 8 **Learning-rate transfer across model scales.** Results across the 255M, 476M, and 1B validation scales. The control run type indicates final loss from the optimal configuration found in the initial smaller-scale reference sweep. The predicted, optimal LR results in a better loss than both this control and all other configurations at the same scale (sweep run type), for all model sizes.

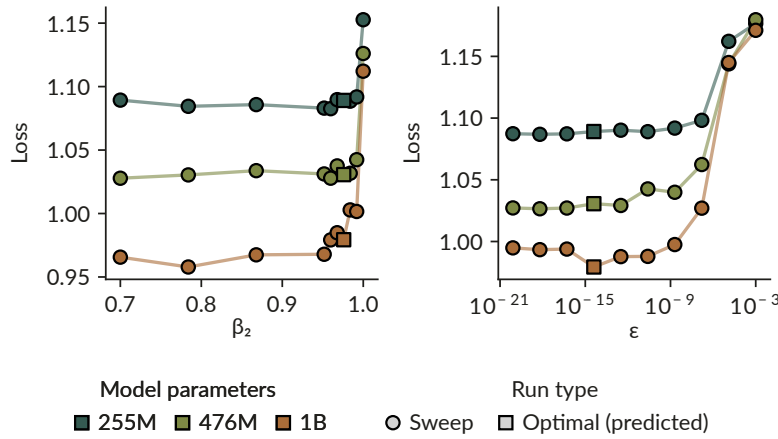


Figure 9 **Adam β_2 and ϵ transfer.** Results across the same scales as Figure 8.

That validation is a fairly unforgiving test. If the transferred learning rate were merely close by accident, it would be surprising for it to land correctly across all three validation scales, but the prediction remains well centered at each one. For DNA, that is a pretty cool result. Prior biology foundation-model work has used μP -style transfer, but we are not aware of a DNA result showing that a more inclusive framework like Complete(d) works across token horizon and batch size, which are the axes we keep leaning on later in ad hoc runs across epochs. The same is mostly true for the other optimizer hyperparameters too, although Adam β_2 shows some signs of being a bit aggressive at the largest scale. [Supplementary Figure S1](#) makes the same point across CDS, upstream, and downstream sequence, with no qualitative difference in transfer behavior across region types. That gives us enough confidence that the following parameter-scaling runs are at least close to optimally configured.

²³Here, “validation loss” is best understood as a training-loss-like monitoring statistic computed on a fixed set of human training sequences, rather than a conventional estimate on held-out data. We have not yet found a satisfactory way to construct clean held-out genomic splits: genomes are phylogenetically correlated, and identifying orthologous non-coding regions by sequence alignment is difficult. See [issue #8](#) for split experiments and a broader discussion of why raw perplexity may not reliably track VEP performance. The meaning of lowercase differs between training and validation: during training it marks repetitive bases, whereas in these validation

2.3. Parameter scaling

Before asking whether better validation loss²³ translates into better VEP performance, we first needed to check whether validation loss scaled the way it should. The parameter sweep uses the same training recipe at each model size, with all hyperparameters set by the transfer heuristic above, and then asks whether the resulting losses fit a Kaplan-style scaling law well (they do).²⁴ Despite this being a simple experiment conceptually, actually getting there took months — fitting the hyperparameter transfer heuristic, running the validation experiments, and training the 4B model, which alone took about three weeks to finish. The final sweep spans 8 model sizes from 46M to 4B parameters, each trained on ~84B tokens, for ~4.3e21 FLOPs across the sweep. That puts it on par with canonical scaling-law studies in language modeling, e.g. its ~2.1e21 FLOP 4B run matches the compute Hugging Face used at that exact model scale in their data-constrained scaling work.²⁵

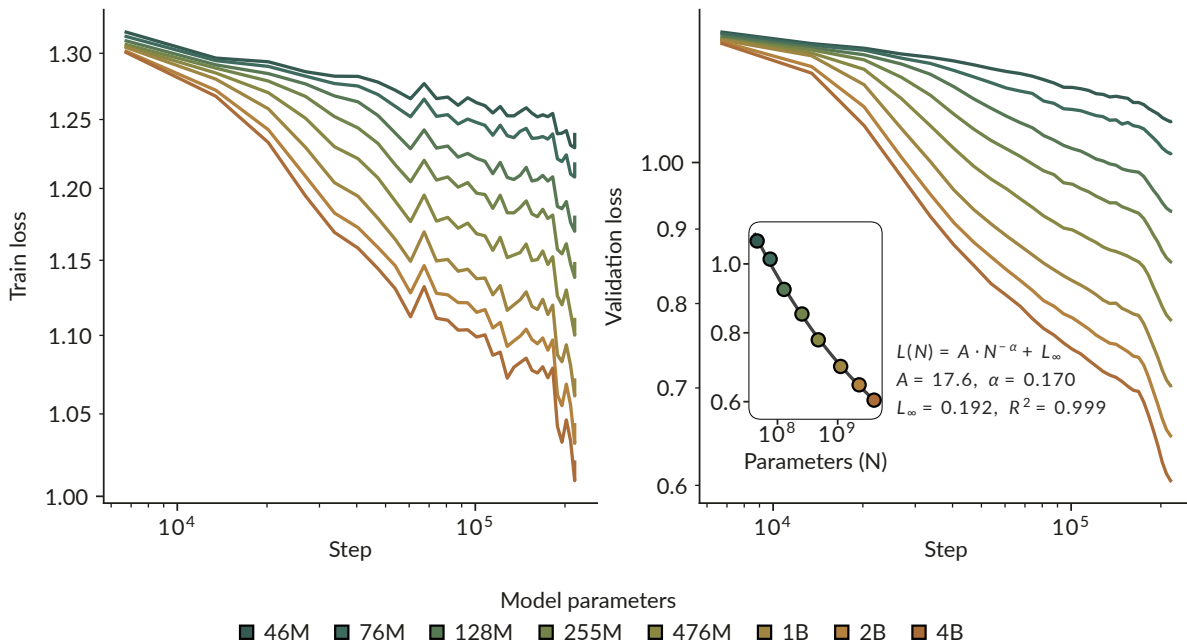


Figure 10 **Loss scaling across model sizes.** Training and validation loss for eight models from 46M to 4B parameters, each trained on ~84B tokens; the inset shows a Kaplan power-law fit to final validation loss. The result is about as tidy as we could hope for. Training is stable at every scale, and both training and validation loss decrease monotonically and predictably, as shown in Figure 10. We use WSD learning-rate schedules with 10% warmup and 20% decay, which causes the visible drop in both losses over the final 20% of tokens. Most importantly, the sweep gives a high-quality Kaplan scaling-law fit ($R^2=0.999$), which makes the next question much better posed. Does lower validation loss actually correlate with better downstream VEP performance?

2.4. Downstream performance

The final sweep shows a mostly consistent relationship between parameter count and downstream VEP performance. When zero-shot LLR and linear probes are evaluated on identical variants, performance improves with scale for most variant types. The clearest exception is Mendelian missense: zero-shot LLR peaks at 128M param-

sets it marks non-conserved bases; in both cases lowercase positions receive 1% of the standard loss weight. In validation, this acts as a heuristic that emphasizes conserved sequence, but we did not independently design or validate it as a model-selection objective. Because this statistic informed the Vizier reference sweep and the choice of the 1B model, those decisions may not be optimal under a different validation objective. We therefore use validation loss descriptively for like-for-like comparisons and trend analysis, not as an unbiased estimate of performance on unseen sequence; the biological conclusions rely primarily on downstream VEP evaluation.

²⁴This follows the empirical scaling-law setup from Kaplan et al., where model loss is fit as a predictable function of model size, data, and compute.

²⁵See Fig. 4 of Muennighoff et al., “Scaling Data-Constrained Language Models” (NeurIPS 2023).

²⁶Non-monotonic likelihood-based zero-shot variant-effect performance with increasing model scale has previously been observed in other settings. Gordon et al. report that ESM-2 performance on protein deep-mutational-scanning benchmarks degrades beyond

ters and then deteriorates, even as linear-probe performance continues to improve.²⁶ This is not a general failure on missense variants—the SGE missense benchmark improves with scale under both scoring protocols.

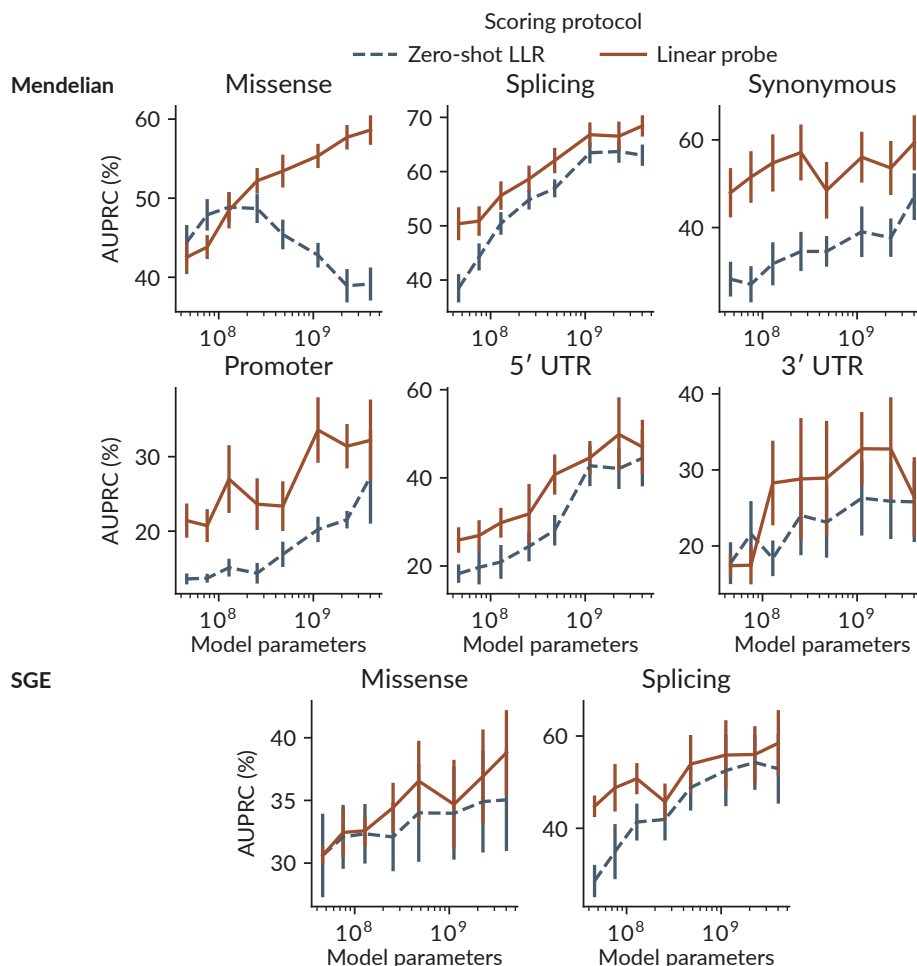


Figure 11 **VEP performance across model scale.** Zero-shot LLR and a linear probe are compared on identical variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

Plotting the same results against matched-region validation log-likelihood gives the same picture. For all other combinations of variant type and scoring protocol, better validation log-likelihood is associated with better downstream performance. Zero-shot Mendelian missense again points in the opposite direction: performance declines even as validation log-likelihood improves.

an intermediate model size and show that performance depends on the likelihood assigned to the wild-type sequence. [Pugh et al.](#) similarly report plateaus or regressions for larger protein language models under standard likelihood-based scoring. These studies concern protein models and different evaluation regimes; we do not know whether our Mendelian missense result has the same cause.

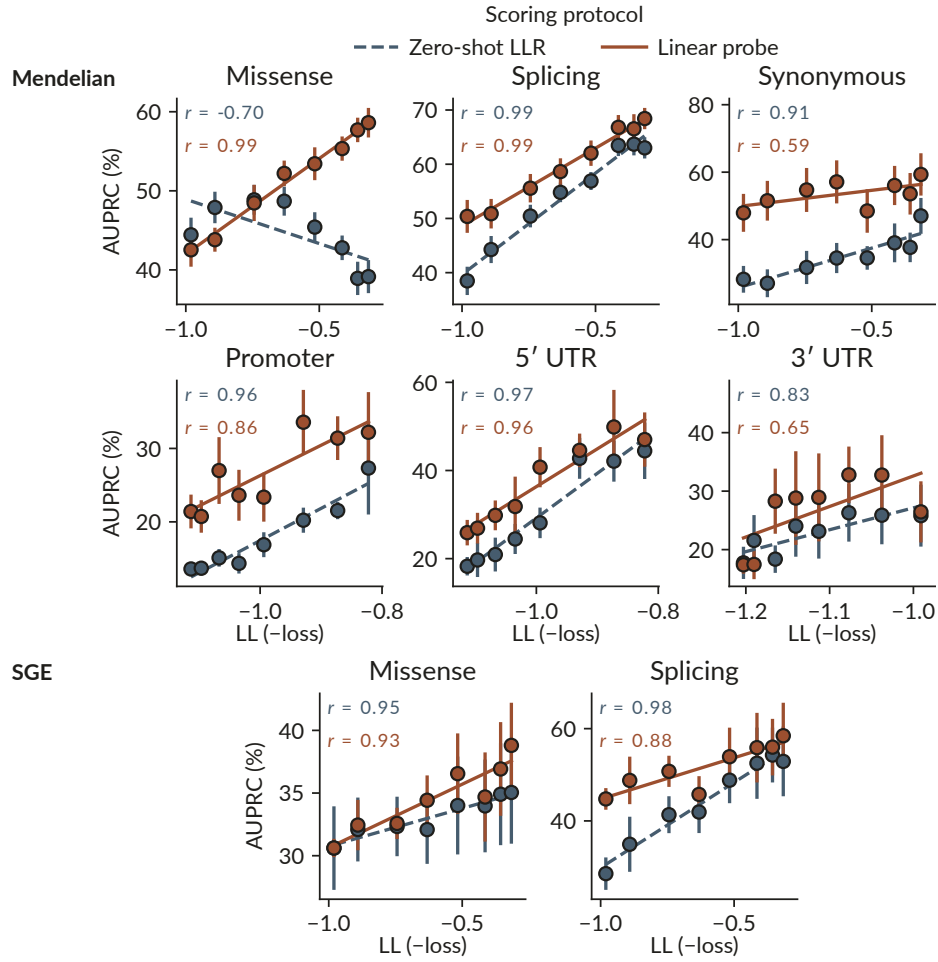


Figure 12 **VEP performance versus validation log-likelihood**. Matched-region LL (shown as $-\text{loss}$) across the eight parameter-scaling endpoints. Performance is measured as chromosome-weighted AUPRC. Lines are least-squares fits, and r denotes Pearson correlation; error bars denote SE.

This divergence is not unique to MarinDNA. On the same Mendelian missense benchmark, Evo 2 also shows improving linear-probe performance alongside declining zero-shot LLR performance as model size increases. For now, this should be interpreted as a recurring pattern for Mendelian missense across these two model families—not as evidence that zero-shot readouts generally deteriorate with scale. It also cautions against using zero-shot LLR alone to judge whether scaling has improved the learned representations for this task.

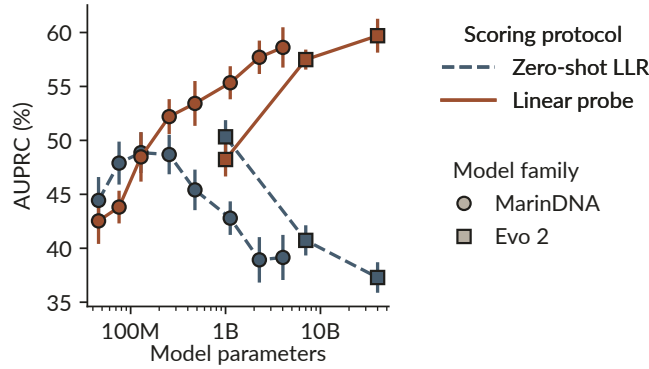


Figure 13 **Missense readouts across model scale.** MarinDNA and Evo 2 are compared using zero-shot LLR and a linear probe on identical Mendelian variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

2.5. Later mixture experiments

At this point we move away from theoretically grounded, compute-constrained methods. The later experiments still rely on the transfer heuristics above, since we need learning rates and other hyperparameters for runs with very different token horizons. But the actual optimization problem becomes much more ad hoc — we start changing mixture constituents, epoch them freely, and see whether in-flight changes can compensate for observed performance gaps.

We standardized these experiments on a 1B-parameter model.²⁷

Our starting point was a uniform mixture of CDS, upstream, and downstream sequence. These were the three region datasets we could initially construct consistently across many species from standard genome annotations. The scaling study had instead sampled these regions in proportion to dataset size. Echoing the earlier mixture experiments, that recipe produced good CDS performance but left a large gap on the less abundant upstream tasks. We therefore switched to uniform weighting so that each functional region received meaningful exposure.

We use the internal labels m5.1, m1.3, and m3.3 for the three model-mixture lineages compared below. In this naming scheme, m denotes a mixture lineage, the leading number identifies the mixture strategy, and the suffix identifies a continuation within that lineage. m5.1 is the staged lineage: we first trained its 1B model for ~104B tokens on the uniform three-region mixture. Alignment projection then made it possible to turn human ncRNA and enhancer annotations into comparable multi-species training datasets. Once those data became available, we added them to form a uniform five-region mixture and continued training the same model for another ~62B tokens. We compare this staged history with two controls, m1.3 and m3.3, trained on five-region mixtures from the beginning.

²⁷ At the time, 1B had reached a good level of zero-shot performance under our then-current evaluation. We had not established it as the optimal model size. The later linear-probe results change this judgment most: they continued to improve with scale even where zero-shot Mendelian missense did not, making a larger model a more compelling choice in retrospect.

CONTINUED-TRAINING RECIPES

What training data did each recipe see?

Each lane spans the same 166B-token window; a stage boundary appears only when the data mixture changes.

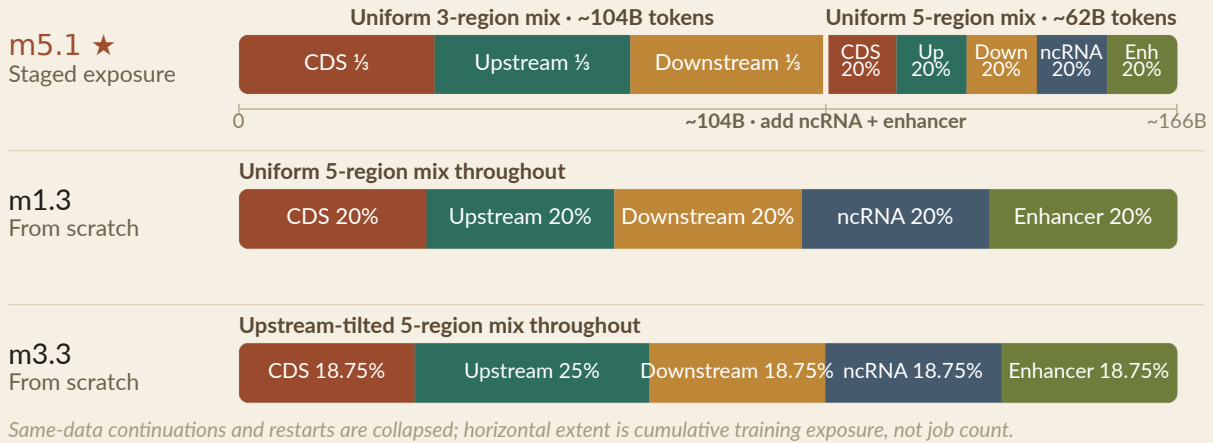


Figure 14 **Training-data exposure histories.** Three recipes are compared. m5.1 trains for approximately 104B tokens on a uniform three-region mixture before adding ncRNA and enhancer data for approximately 62B tokens. The de novo m1.3 and m3.3 controls keep fixed five-region mixtures over the same displayed token horizon; m3.3 gives upstream sequence 25% weight and each other region 18.75%.

For m5.1, the first evaluated checkpoint after adding ncRNA and enhancer data shows gains in variant-effect performance in the corresponding ncRNA and distal subsets. Across all eight subsets, m5.1 ultimately finishes with the highest macro-average AUPRC under both zero-shot LLR and the linear probe, although the linear-probe trajectories are visibly noisier.²⁸ Its advantage is broad but not universal: the lineages trained on five regions from the beginning retain stronger endpoints for some distal and ncRNA subsets. m5.1’s strong endpoint raises the possibility that exposure order matters: learning first from the three-region mixture and introducing ncRNA and enhancer data later may be more effective than training on all five regions from the beginning, though this remains uncertain and requires further investigation.

The corresponding linear-probe trajectories appear in [Supplementary Figure S2](#).

²⁸The curves shown here use a separate probe trained within each variant subset. In a [separate 255M analysis on one held-out chromosome](#), training one probe across all subsets improved the AUPRC point estimate on several data-starved subsets, including ncRNA and distal, while hurting stronger or more specialized subsets. This makes limited labeled data one plausible contributor to the noise, but that analysis did not directly test the checkpoint-to-checkpoint variability in these 1B lineage curves.

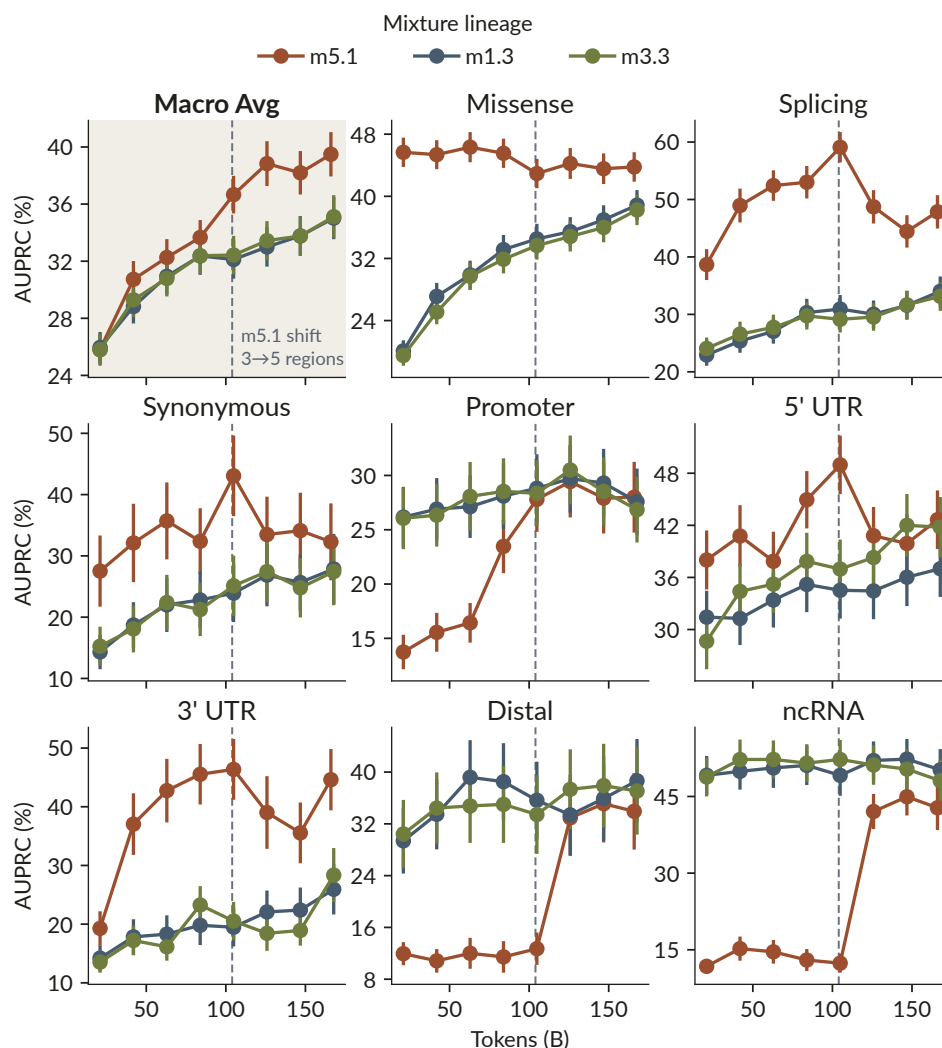


Figure 15 **Zero-shot mixture-lineage trajectories.** Mendelian AUPRC versus training tokens for three model-mixture lineages. Error bars denote SE.

2.6. Leaderboard scores

The result of the previous mixture experiments is m5.1, a 1B GPT-style model evaluated alongside other models on our [live Mendelian VEP leaderboard](#), where we continue to add experimental runs and baselines. In the zero-shot snapshot shown here, m5.1 comes out slightly ahead of Evo 2 40B. Its advantage is considerably larger under linear probing.

m5.1 was trained on 166B tokens ($\sim 1.1e21$ FLOPs), compared with 9.3T tokens ($\sim 2.25e24$ FLOPs) for Evo 2 40B. At their native context lengths on the same GH200, m5.1 scores one million variants in about 41 minutes, compared with roughly 66 days for Evo 2 40B—a roughly 2,330 $\times$ throughput advantage.²⁹

²⁹This benchmark measures steady-state scoring with forward and reverse-complement passes and embeddings enabled. m5.1 uses a 256-token context, while Evo 2 40B uses an 8,192-token context, so this measures as-deployed throughput rather than same-context or per-token efficiency. See [issue #354](#) for the full methodology and results.

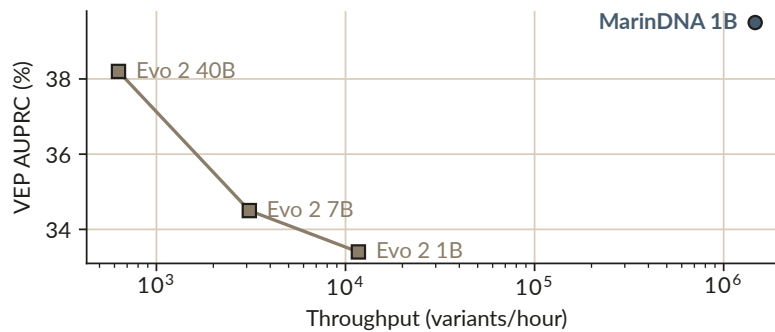


Figure 16 **Variant-effect performance versus as-deployed inference throughput.** Zero-shot Mendelian macro-average AUPRC versus variants scored per hour on one GH200. Each model is evaluated at its native context length; throughput includes forward- and reverse-complement scoring and embedding output.

Most notably, m5.1 closes Evo 2’s main gap on distal variants, outperforming Evo 2 40B there under both readouts. The improvement is not uniform, however: Evo 2 40B remains ahead on splicing under both readouts, as well as on promoter and synonymous variants in the zero-shot evaluation and missense variants under linear probing.

The [current leaderboard](#) also suggests there is considerable headroom: although m5.1 leads on Macro Avg among MarinDNA models, another MarinDNA run has a higher point estimate in seven of the eight displayed subsets, with m5.1 leading only on ncRNA. Several of these winners are region specialists, suggesting that further mixture refinement could recover more of their complementary strengths.

On the broader zero-shot leaderboard, m5.1 remains slightly behind AlphaGenome and substantially behind GPN-Star.³⁰ These are different model families: AlphaGenome learns from functional-genomics supervision, while GPN-Star uses whole-genome alignments. Further improvements to the alignment-free recipe may narrow the gap to GPN-Star, but it is not clear that they will eliminate it. Some of the remaining difference may reflect information that an alignment-free model cannot recover from unaligned sequence alone ([research question #397](#)).

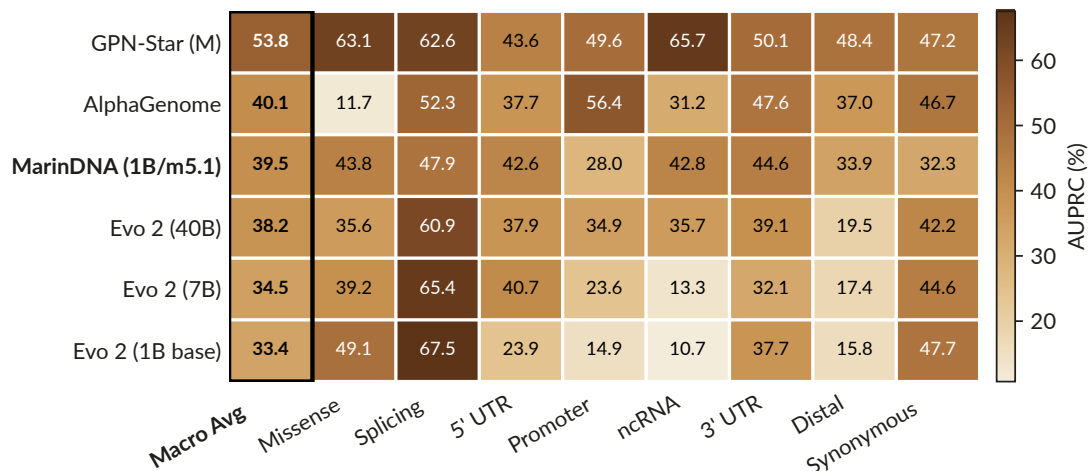


Figure 17 **Zero-shot Mendelian leaderboard.**

³⁰Figure 17 uses LLR for MarinDNA and Evo 2, calibrated LLR (cLLR) for GPN-Star, and the maximum L2 REF/ALT prediction score for AlphaGenome.

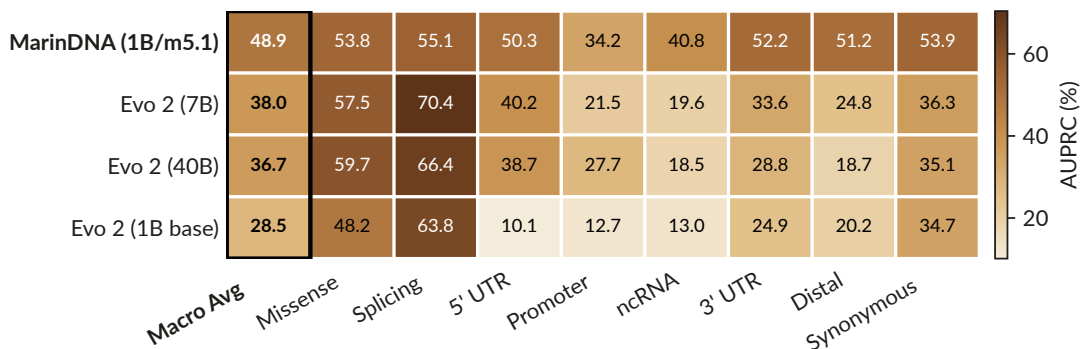


Figure 18 **Linear-probe Mendelian leaderboard**. AUPRC is a sample-size-weighted average across chromosomes and is not directly comparable with Figure 17. GPN-Star and AlphaGenome are absent because no compatible probe results are available here, not because of their performance.

3. Discussion

MarinDNA shows that a conventional decoder-only Transformer can become a strong genomic sequence model when data construction and optimization are treated as primary modeling choices. The final 1B model establishes the attainable performance and efficiency, while the mixture and hyperparameter-transfer experiments explain how that result was reached.

The data-mixture results indicate that proportional sampling is not a neutral default when genomic region datasets differ substantially in size. In the upstream/CDS experiment, proportional pooling behaved similarly to CDS-only training, whereas equal sampling retained useful performance across both region types. The later m5.1 lineage extended this principle by introducing alignment-projected ncRNA and enhancer data and finished with the strongest macro-average endpoint among the three frozen mixture lineages. This result supports explicit control of region exposure, but it does not establish that staged exposure is intrinsically better than de novo five-region training because the lineages are not a controlled curriculum ablation.

The hyperparameter experiments provide complementary evidence that optimization settings can be transferred across model size, batch size, and token horizon in this regime. The transferred learning rate was the best observed tested value at each of three target scales, and the resulting parameter ladder trained stably through 4B parameters. The transfer rule nevertheless inherits assumptions from a text-model recipe, including its token-horizon exponent and model-geometry heuristic, and the present experiments do not establish that those choices are globally optimal for DNA.

On the frozen Mendelian evaluation, the final MarinDNA model is statistically competitive with Evo 2 40B rather than demonstrably superior to it. Point estimates differ by consequence class and readout: MarinDNA closes much of the distal-variant gap, while Evo 2 retains advantages on several coding and splicing subsets. GPN-Star and AlphaGenome remain stronger in the broader zero-shot comparison, but they use alignment-derived information and functional-genomics supervision, respectively, and therefore bound rather than directly match the alignment-free setting studied here.

The compute and throughput comparisons require similar care. The training-FLOP ratio uses reported total training compute, whereas the throughput measurement compares steady-state scoring on the same GH200 at each model’s native context length and includes forward- and reverse-complement passes plus embedding output. It therefore measures the cost of deploying the evaluated models for this workload, not architecture-only efficiency at matched context length or matched tokens processed.

Several limitations constrain the biological interpretation. The models use a short 255-base context chosen for functional-element evaluation, so these experiments do not test long-range regulatory interactions or genome-scale sequence generation. Human variant-effect data provide the strongest available evaluation, but performance in humans does not by itself establish transfer to non-model organisms, which is an important motivation for alignment-free models. The weighted validation loss is computed on fixed human sequences related to the training

distribution rather than a clean phylogenetically independent holdout, so it is useful for controlled comparisons but not an unbiased estimate of genomic generalization. Finally, the zero-shot Mendelian missense regression with scale shows that likelihood-based readouts can diverge from information retained in frozen representations. Neither scaling nor optimization appears exhausted: linear-probe performance continued to improve through the largest evaluated model, and the loss-scaling fit showed no clear saturation over the tested range.

4. Methods

4.1. Study design and frozen analysis snapshot

This manuscript reports the model, evaluation, and figure snapshot used for the published MarinDNA article rather than a continuously updated leaderboard. The mechanical manuscript baseline and its 20 SVG assets were fixed at MarinDNA commit [3b608d39b41c2330636ec647dbb25d26b0895187](#), whose content and figures were converted from project commit [d8c4803cbbbffafb24890cd0c75134d78368d55c](#). No new training run or evaluation dataset was introduced for the editorial conversion. The live leaderboard is treated as a separate, evolving resource.

4.2. Training data

Training examples were drawn from functional regions across multiple animal genomes. The initial datasets contained coding sequences (CDS), sequence upstream of annotated genes, and sequence downstream of CDS ends. The downstream category denotes the 256 bases immediately after an annotated CDS end rather than annotated 3' UTR intervals. Later training mixtures added functional ncRNA regions and ENCODE V4 non-promoter candidate cis-regulatory elements, with human annotations projected to other species by sequence alignment. All genomic coordinates inside MarinDNA use 0-based, half-open intervals, with conversion to external coordinate conventions only at tool boundaries. The later recipe represents each example with 255 DNA bases and a beginning-of-sequence token, giving a 256-token model input. Soft-masked repetitive bases receive 1% of the standard training-loss weight. Training-data revisions and per-region token counts for every retained experiment will be enumerated in the supplementary provenance table.

4.3. Model and training objective

MarinDNA uses the Qwen3 decoder-only Transformer architecture with a causal next-token objective and nucleotide-level DNA tokenization. The architecture is intentionally conventional: the experiments vary training data, optimization, and parameter scale without introducing a genomics-specific sequence operator. The final m5.1 model has approximately 1B parameters and a 255-base context. It was first trained for approximately 104B tokens on a uniform mixture of CDS, upstream, and downstream data, then continued for approximately 62B tokens after ncRNA and enhancer data were added to form a uniform five-region mixture. The frozen internal checkpoint is mix-v0.9-p1B-i24-exp135-m5.1-step-59158 at GCS path `gs://marindna-public-model-revision/`. An immutable public model revision will be recorded in Section 5 before preprint release.

4.4. Hyperparameter calibration and transfer

A Bayesian reference sweep using Vizier trained approximately 25M-parameter models for 2.5B tokens with a 16k-token batch. The sweep tuned initialization scale, AdamH and Adam peak learning rates, β_1 , β_2 , ϵ , gradient clipping, and z-loss against the weighted validation-loss statistic described below. A fixed geometry rule maps hidden width to the number of layers, MLP width, attention heads, and key-value heads. A second fixed rule transfers the reference optimizer settings to a target batch size and token horizon. Relative to reference batch size B_0 and horizon T_0 , the AdamH learning rate is multiplied by $\sqrt{(B/B_0)(T_0/T)^{0.3}}$. The Adam learning rate is multiplied by $\sqrt{(r/r_0)}$, ϵ by $\sqrt{(r_0/r)}$, and β_2 is exponentiated by B/B_0 subject to configured bounds, where $r/r_0 = (B/T_0)/(B_0/T)$. Initialization scale, β_1 , gradient clipping, and z-loss are reused without transformation. Both learning rates use linear warmup over the first 10% of steps, remain at their peak through 80% of training, and decay linearly to zero over the final 20%. Target-scale validation used 255M-, 476M-, and 1B-parameter models with four times the token horizon and one quarter of the reference batch size. “Best observed” refers to the lowest final validation loss among the tested target-scale settings and does not imply a global optimum.

4.5. Parameter-scaling experiment and validation loss

The parameter ladder contains models of approximately 46M, 76M, 128M, 255M, 476M, 1B, 2B, and 4B parameters. Each model was trained for approximately 84.77B tokens with the transferred optimization recipe. The final validation losses were fit with a Kaplan-style power law as a function of parameter count. Training and validation losses weight soft-masked or non-conserved lowercase bases at 1% of the standard token weight. The validation statistic is computed on fixed human sequences related to the training distribution and is used for controlled comparisons and model selection. It is not interpreted as an unbiased estimate on a phylogenetically independent held-out genome set.

4.6. Variant-effect datasets

The Mendelian benchmark compares pathogenic variants with non-rare gnomAD controls matched within consequence class. Relative to the TraitGym design, the minimum control allele frequency is 0.1%, the consequence set includes missense and splicing variants, and matching includes potential confounders such as transcription-start-site distance and exon distance where applicable. The frozen benchmark revision is [4aed58e50c5dea0b878a665007af2ef9e5108e9f](#). The saturation-genome-editing benchmark is the v3 labeled build at [225d3d1ea32a4af547891b13c33b5e92a5aae849](#). It retains variants labeled abnormal or normal by ClinGen/ExCALIBR-calibrated assay thresholds and groups them into missense and splicing subsets. Variant contexts are extracted from the Ensembl release 115 GRCh38 soft-masked primary assembly. The evaluation pipeline uses the development split for model iteration; the chromosome-disjoint test split remains reserved for the final locked evaluation.

4.7. Zero-shot likelihood scoring

For each allele, the causal model scores the variant-centered sequence on the forward strand and its reverse complement. The raw log-likelihood ratio is defined as $\log P(\text{ALT}) - \log P(\text{REF})$. For Mendelian and SGE evaluation, deleteriousness is oriented as the negative of the mean forward- and reverse-complement log-likelihood ratio, so larger scores indicate a larger likelihood penalty for the alternate allele. Forward and reverse-complement scores are retained separately, but their mean is the default MarinDNA and Evo 2 score used in the headline comparison. GPN-Star is represented by its calibrated likelihood-ratio score, whereas AlphaGenome is represented by its maximum L2 reference/alternate prediction score; these scores are not interchangeable measurements of model likelihood.

4.8. Frozen-embedding linear probes

The same forward and reverse-complement passes used for likelihood scoring also produce last-layer hidden states for the reference and alternate alleles. Hidden states are mean-pooled across the entire DNA window with special tokens excluded, accumulated and strand-averaged in float32, and stored in float16. For directional Mendelian and SGE labels, the probe feature concatenates the reference embedding with the alternate-minus-reference embedding after upcasting to float32. A separate StandardScaler plus L2-regularized logistic-regression pipeline is fit within each consequence subset. Predictions use leave-one-chromosome-out cross-validation, with the regularization strength retuned inside each outer fold by chromosome-grouped cross-validation over a fixed 17-point grid from $1e-12$ to $1e4$. Probe and zero-shot metrics are calculated on identical variant rows.

4.9. Metrics and uncertainty

Performance is summarized by area under the precision-recall curve (AUPRC). For matched Mendelian zero-shot analyses, AUPRC is computed within consequence subset and uncertainty is estimated by resampling matched groups with replacement, thereby preserving the one-to-many matched structure. For probe analyses, AUPRC is first calculated per chromosome and then weighted by the number of contributing variants; uncertainty is estimated by a chromosome-cluster bootstrap. SGE AUPRC is computed separately within each MaveDB accession and consequence subset before macro-averaging because scores are not comparable across studies. Unless a caption states otherwise, error bars denote ± 1 bootstrap standard error. Paired model comparisons on matched variants resample matched groups once per iteration and recompute both AUPRC values on the same resample. The comparison reports the observed AUPRC difference, its bootstrap standard error, the 2.5th and 97.5th percentiles of the paired difference distribution, and a two-sided bootstrap p-value. Comparative language

in the title, abstract, Results, and Discussion follows this paired uncertainty rather than the ordering of point estimates alone.

4.10. Training compute and inference throughput

Training compute is reported as model-training FLOPs for the complete runs. The final m5.1 model used approximately $1.1\text{e}21$ FLOPs over 166B tokens, whereas Evo 2 40B reports approximately $2.25\text{e}24$ FLOPs over 9.3T tokens. Inference throughput was measured on one NVIDIA GH200 for steady-state scoring with forward- and reverse-complement passes and embedding output enabled. The workload scores one million variants at each model’s native context length: 256 tokens for m5.1 and 8,192 tokens for Evo 2 40B. The resulting comparison is an as-deployed workload measurement rather than a same-context, same-batch, or per-token architecture benchmark.

5. Data, Code, Models, and Live Resources

The [MarinDNA repository](#) contains the data pipelines, evaluation implementation, manuscript source, and tracked investigation scripts. The mechanical manuscript and frozen figure baseline is available at commit [3b608d39b41c2330636ec647dbb25d26b0895187](#). The final preprint revision will identify commit-pinned training, dataset-construction, evaluation, and plotting code for each retained result.

The frozen Mendelian benchmark is [bolinas-dna/evals_mendelian_traits](#) at revision [4aed58e](#). The frozen saturation-genome-editing benchmark is [bolinas-dna/evals_sge](#) at revision [225d3d1](#). Training datasets, evaluation datasets, and the released model are grouped in the [MarinDNA Hugging Face collection](#). The exact public model and training-dataset revisions will be pinned here before the preprint is released.

The [MarinDNA leaderboard](#) is an evolving project resource. Its contents may change as models and baselines are added and must not be treated as the frozen comparison reported in this manuscript. The committed manuscript figures and their recorded input revisions define the archival result snapshot.

Optional interactive resources include:

- [Interactive sequence explorer](#).
- [Model inference and BRCA1 variant-effect prediction notebook](#).

6. Open Development and Provenance

MarinDNA was developed through public, issue-tracked experiments in which hypotheses, configurations, intermediate results, and negative findings were recorded as the work progressed. The issue history provides development provenance but is not required scientific exposition: the manuscript states the methods, evidence, and limitations needed to interpret its claims. Decisions and substantial editorial changes for this frozen preprint are tracked in [issue #449](#).

Key result families retain the following development records, which will be expanded with exact comments, code commits, checkpoints, dataset revisions, and figure inputs in the supplementary provenance table:

- Data specialists and mixture balance: [#21](#), [#27](#), and [#13](#).
- Manual scaling failure and the need for transfer: [#57](#).
- Validation-loss design and interpretation: [#8](#).
- Frozen-embedding probe development: [#369](#) and the evaluation-pipeline records it references.
- Throughput measurement: [#354](#).

Issue links are provided to make the branching research history inspectable; scholarly claims about prior work use bibliographic citations, and claims about MarinDNA are supported by the manuscript’s figures, Methods, and frozen artifacts.

7. Acknowledgements

We thank Isaac Hodes, Yael Elmatad, David Hall, Will Held, and Jeff Hammerbacher, as well as others across the Open Athena and Marin communities, for thoughtful discussions. We also thank the Google TPU Research Cloud (TRC) program for providing compute resources.

8. Author Contributions

TODO: Complete after the author list and order are agreed.

9. Competing Interests

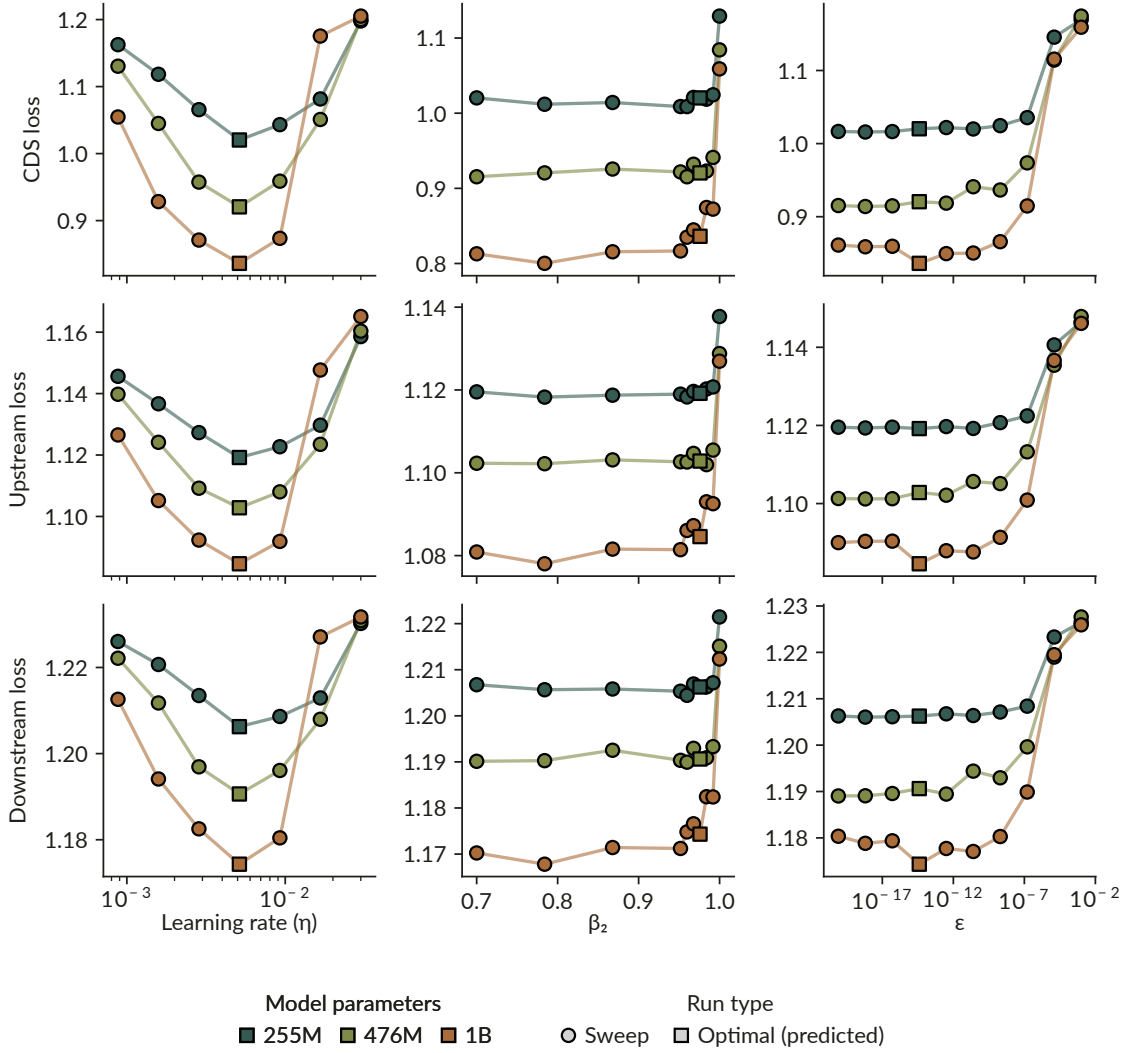
TODO: Obtain and record declarations from all authors before public posting.

10. Funding and Compute Resources

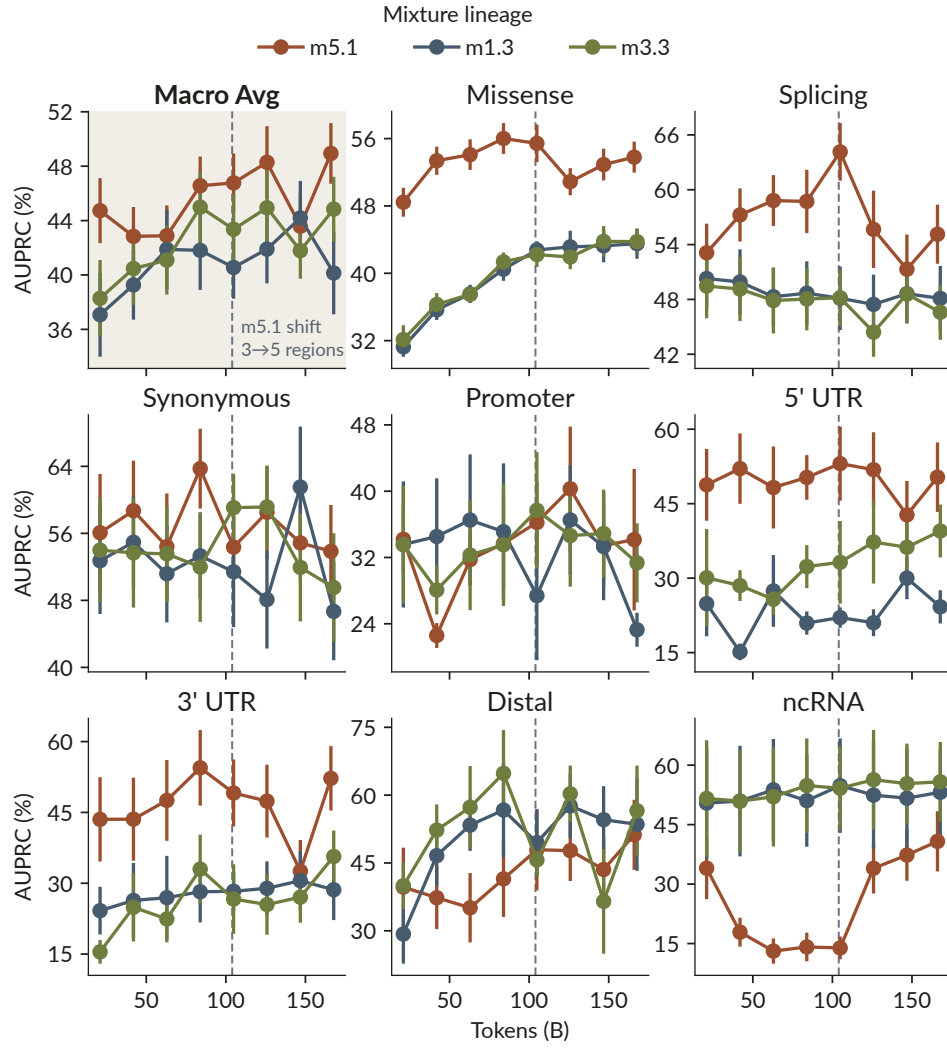
Training used compute resources provided by the Google TPU Research Cloud program. Additional funding and resource disclosures will be completed before public posting.

Supplementary Information

The figures below were hidden behind disclosure controls in the published blog presentation and are included here as supplementary figures.



Supplementary Figure S1 **Region-specific hyperparameter transfer**. Validation loss as a function of learning rate, β_2 , and ϵ , evaluated separately for CDS, upstream, and downstream regions.



Supplementary Figure S2 **Linear-probe mixture-lineage trajectories**. Linear-probe Mendelian AUPRC (chromosome-averaged) versus training tokens for three model-mixture lineages. Error bars denote SE.